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DRILL W05

④ Linear Regression:

1) Data: $X = \begin{bmatrix} 1 & x_{11} & x_{12} & \dots & x_{1(D-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{N1} & x_{N2} & \dots & x_{N(D-1)} \end{bmatrix}$; $t = \begin{bmatrix} t_1 \\ t_2 \\ \vdots \\ t_N \end{bmatrix}$
(Input)

$\Rightarrow$ Dimension: $X: N \times D$; $t: N \times 1$. $\rightarrow t = \underbrace{h(x)}_{\text{unknown function}} + \underbrace{\epsilon}_{\text{Noise, } \sim N(0, \sigma^2)}$ (1)

2) Model: $\hat{y}(x, w) = w_0 \cdot x_0 + w_1 \cdot x_1 + \dots + w_{D-1} \cdot x_{D-1}$
 $= w^T x$ (2)

$\rightarrow$ Output: $w = [w_0, w_1, \dots, w_{D-1}]$ (~~coefficients~~ coefficients of Model)

(1), (2) $\Rightarrow t \sim N(t | \hat{y}(x, w), \beta^{-1})$ ($\beta^{-1} = \sigma^2$)
 $\Rightarrow p(t | x, w, \beta)$ (likelihood function)

$= \frac{1}{\sqrt{2\pi}} \cdot \exp\left(-\frac{\beta}{2} [t - \hat{y}(x, w)]^2\right)$
 $= \frac{\beta^{1/2}}{(2\pi)^{1/2}} \cdot \exp\left(-\frac{\beta}{2} [t - \hat{y}(x, w)]^2\right)$ (3)

Probability: $P(x | \theta)$
 $\rightarrow x$ is variable, $\theta = \text{const}$
Likelihood: $L(\theta | x)$
 θ is variable, $x = \text{const}$ (dataset)

Joint probability of occurrence for target values in $t = [t_1, t_2, \dots, t_N]^T$:
 $p(t | X, w, \beta) = \prod_{n=1}^N p(t_n | X, w, \beta)$ (4)

$\ell(w, \beta) \triangleq -\log p(t | X, w, \beta) = -\sum_{n=1}^N \log p(t_n | X, w, \beta)$

From (3) $\Rightarrow$
 $L(w, \beta) = -\sum_{n=1}^N \log \left(\frac{\beta^{1/2}}{(2\pi)^{1/2}} \exp\left(-\frac{\beta}{2} [t_n - \hat{y}_n(x, w)]^2\right) \right)$
 $= -\sum_{n=1}^N \left(\log\left(\frac{\beta^{1/2}}{(2\pi)^{1/2}}\right) + \log\left(\exp\left(-\frac{\beta}{2} [t_n - \hat{y}_n(x, w)]^2\right)\right) \right)$
 $= -\sum_{n=1}^N \left(\frac{1}{2} \log\left(\frac{\beta}{2\pi}\right) - \frac{\beta}{2} [t_n - \hat{y}_n(x, w)]^2 \right)$
 $= -\sum_{n=1}^N \left(\frac{1}{2} \log \beta - \frac{1}{2} \log(2\pi) - \frac{\beta}{2} [t_n - \hat{y}_n(x, w)]^2 \right)$
 $\hat{y}_n(x, w) = w^T x_n(x) \Rightarrow -\frac{N}{2} \log \beta + \frac{N}{2} \log(2\pi) + \beta \left(\frac{1}{2} \sum_{n=1}^N [t_n - w^T x_n]^2 \right)$
 $E_\beta(w)$

→ Maximize likelihood $(p(t|X, w, \beta)) \Leftrightarrow$ Minimize $L(w, \beta) \Leftrightarrow$ minimize $E_D(w)$ w.r.t w .

$$\Rightarrow \frac{dE_D}{dw} = \frac{1}{2} \sum_{n=1}^N [w^T x_n - t_n] \cdot x_n^T = 0^T$$

$$\Rightarrow \left(\sum_{n=1}^N x_n x_n^T \right) w - \sum_{n=1}^N t_n x_n^T = 0^T$$

$$\Rightarrow w_{ML} = (X^T X)^{-1} X^T t$$

X : dimensions $N \times D$

$X^T X$: $D \times D$

t : $N \times 1$

w_{ML} : $D \times 1$

⊛ Logistic Regression:

$$P(y|x, w) = \begin{cases} \hat{y} & \text{if } y=1 \\ 1-\hat{y} & \text{if } y=0 \end{cases}$$

$$\Rightarrow p(y|x, w) = \hat{y}^y (1-\hat{y})^{1-y}$$

$$\Rightarrow p(y_n|x_n, w) = \hat{y}_n^{y_n} (1-\hat{y}_n)^{1-y_n}$$

$$\Rightarrow p(t|X, w) = \prod_{n=1}^N \hat{y}_n^{y_n} (1-\hat{y}_n)^{1-y_n}$$

→ Negative Log-Likelihood: $L(w) \triangleq -\log p(t|X, w)$

$$= - \sum_{n=1}^N [y_n \log \hat{y}_n + (1-y_n) \log (1-\hat{y}_n)]$$

Derivative:

$$\Delta w \triangleq \frac{dL(w; x, y)}{dw} = \frac{dL}{dz} \cdot \frac{dz}{dw} = (\hat{y} - y) \cdot \frac{dz}{dw} = (\hat{y} - y) \cdot x$$

⇒ On set of N data points (X, y) : $\Delta w = X^T (\hat{y} - y)$

2nd-Order derivative (Hessian Matrix) of $L(w)$:

$$H \triangleq \nabla \nabla L(w) = \sum_{n=1}^N \hat{y}_n (1-\hat{y}_n) x_n x_n^T$$

$$= X^T R X$$

R is a diagonal matrix with element $R_{nn} = \hat{y}_n (1-\hat{y}_n)$